```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
  ABLATION RUN — batch_size=32
  Random loss baseline: log(32) = 3.466
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Downloading: "https://download.pytorch.org/models/resnet50-0676ba61.pth" to /root/.cache/torch/hub/checkpoints/resnet50-0676ba61.pth
100%|██████████| 97.8M/97.8M [00:00<00:00, 163MB/s]
model.safetensors: 100% ████████████████ 268M/268M [00:02<00:00, 120MB/s]
  Epoch 01/08 | train 1.7631 | val 1.3874 ← best
  Epoch 02/08 | train 1.0380 | val 1.2473 ← best
  Epoch 03/08 | train 0.7031 | val 1.2639
  Epoch 04/08 | train 0.4734 | val 1.1973 ← best
  Epoch 05/08 | train 0.3061 | val 1.2211
  Epoch 06/08 | train 0.2131 | val 1.1923 ← best
  Epoch 07/08 | train 0.1490 | val 1.1798 ← best
/content/mini-clip-DLAI/src/train.py:136: FutureWarning: You are using `torch.load` with `weights_only=False` (the current default value), which uses the def
  checkpoint = torch.load(load_path, map_location=device)
  Epoch 08/08 | train 0.1214 | val 1.1869

  batch_size=32
──────────────────────────────────────────
  Metric                Score
──────────────────────────────────────────
  Text → Image
    R@1       20.00%  ████
    R@5       45.80%  █████████
    R@10      59.30%  ███████████
  Image → Text
    R@1       18.60%  ███
    R@5       46.60%  █████████
    R@10      61.50%  ████████████
──────────────────────────────────────────
```

```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
  ABLATION RUN — batch_size=64
  Random loss baseline: log(64) = 4.159
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
  Epoch 01/08 | train 2.2580 | val 1.8177 ← best
  Epoch 02/08 | train 1.3108 | val 1.5859 ← best
  Epoch 03/08 | train 0.8693 | val 1.5420 ← best
  Epoch 04/08 | train 0.5797 | val 1.5356 ← best
  Epoch 05/08 | train 0.3723 | val 1.4874 ← best
  Epoch 06/08 | train 0.2559 | val 1.4792 ← best
  Epoch 07/08 | train 0.1838 | val 1.4761 ← best
  Epoch 08/08 | train 0.1520 | val 1.4830

  batch_size=64
 ──────────────────────────────────────────
  Metric                 Score
 ──────────────────────────────────────────
  Text → Image
    R@1       21.10%
    R@5       48.20%
    R@10      61.70%
  Image → Text
    R@1       21.60%
    R@5       50.00%
    R@10      62.80%
 ──────────────────────────────────────────
```

```
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
  ABLATION RUN — batch_size=128
  Random loss baseline: log(128) = 4.852
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
  Epoch 01/08 | train 3.0937 | val 2.4340 ← best
  Epoch 02/08 | train 1.8495 | val 2.0583 ← best
  Epoch 03/08 | train 1.2456 | val 1.9938 ← best
  Epoch 04/08 | train 0.8312 | val 2.0185
  Epoch 05/08 | train 0.5723 | val 1.9239 ← best
  Epoch 06/08 | train 0.4175 | val 1.9292
  Epoch 07/08 | train 0.3057 | val 1.8755 ← best
  Epoch 08/08 | train 0.2733 | val 1.8598 ← best

  batch_size=128
──────────────────────────────────────────
  Metric                     Score
──────────────────────────────────────────
  Text → Image
    R@1        21.80%  ████
    R@5        46.90%  █████████
    R@10       62.80%  ████████████
  Image → Text
    R@1        21.20%  ████
    R@5        49.80%  █████████
    R@10       62.30%  ████████████
──────────────────────────────────────────

All ablation runs complete.
```